DEPARTMENT OF DATA ANALYTICS & MATHEMATICAL SCIENCES
JAIN DEEMED-TO-BE UNIVERSITY

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TITLE:

An Innovative Approach to Face Recognition Attendance System- Sunglasses with YOLO V9 Technology

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Introduction

FACIAL RECOGNITION

- The realm of facial recognition technology has witnessed **remarkable advancements** from past times, primarily fueled by the strides made in **deep learning techniques**.
- **Paul Viola and Michael Jones** were two computer vision researchers who proposed **“Rapid Object Detection”** in **2001**
- Since then, the face recognition model and techniques helped in Healthcare, Security, Attendance, Retail (Pay by face), etc

Project Overview

- Design of sunglasses fitted with a micro-HD camera and processor.
- Use YOLOv9 for fast and accurate face detection.
- Automates attendance based on real-time facial recognition.
- Portable, non-intrusive, and efficient for institutions.

Problem Statement

- Existing attendance systems require manual input or fixed cameras.
- Accuracy decreases with face masks, distance, side angles, and poor lighting.
- Need for a hands-free, mobile solution with high precision.
- Objective: Build a compact YOLOv9-based system integrated into eyewear.

Motivation / Significance

- Increasing demand for automated attendance in schools and workplaces.
- Wearable technology allows discreet and convenient monitoring.
- YOLOv9 ensures reliability even for small face detection in real-world settings.
- Enhances productivity and minimizes manual errors.

Data Source & Description

- Dataset consists of 3 facial images per student captured at different angles.
- Images stored locally for training the recognition model.
- Real-time frames captured through integrated sunglasses camera.
- YOLOv9 pre-trained weights used as foundation for fine-tuning.

Data Preprocessing

- Image resizing, normalization, and noise reduction.
- Ensuring consistent lighting and angle variety for robust training.
- Encoding student faces into unique feature vectors.
- Creation of a labeled dataset to map identities accurately.

Exploratory Data Analysis

- Face images analyzed for lighting, angle variation, clarity.
- System tested across different environments: indoor, outdoor, corridor.
- YOLOv9 evaluated for detection speed and bounding box precision.
- Performance remained stable even for partially occluded faces.

System Architecture

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- Sunglasses camera captures real-time video stream.
- YOLOv9 detects faces and extracts key features.
- Processor compares extracted features with stored database.
- Matches are logged automatically as attendance.

Methodology

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- Step 1: Dataset creation and preprocessing.
- Step 2: Fine-tuning YOLOv9 on student face samples.
- Step 3: Building recognition pipeline (detection → extraction → match).
- Step 4: Testing accuracy in real-time video scenarios.

Why YOLOv9?

- High stability and optimized for real-time object detection.
- Performs exceptionally well on small objects like faces.
- Lightweight enough for edge devices such as onboard processors.
- YOLOv10–12 offer marginal improvements but require heavier hardware.

Results & Discussion

- Real-time detection achieved with minimal latency.
- High accuracy for faces at varying distances and angles.
- System reliably logged attendance without manual intervention.
- Demonstrated strong potential for real institutional deployment.

Key Findings

- YOLOv9 provides excellent balance between speed and accuracy.
- Wearable-based detection improves flexibility compared to static cameras.
- Prototype successfully detects and identifies faces in dynamic settings.
- Face recognition pipeline shows high consistency in trials.

Limitations

- Limited student dataset (few images per person).
- Difficulties with heavily occluded faces (mask, scarf, hijab).
- Processor constraints limit ultra-high-resolution video.
- Performance may vary in extreme lighting conditions.

Future Work

- Incorporate thermal detection for improved accuracy.
- Train model with larger datasets including occluded faces.
- Upgrade hardware for faster inference.
- Develop mobile app integration for admin dashboards.

Conclusion

- The YOLOv9 sunglasses system demonstrates innovative use of wearable AI.
- Provides accurate, efficient, and portable attendance automation.
- Strong foundation for future expansions into security and monitoring.
- Represents a practical step toward smart, AI-enabled campuses.


Thank you